

Loop Closure from Two Views: Revisiting PGO for Scalable Trajectory Estimation through Monocular Priors

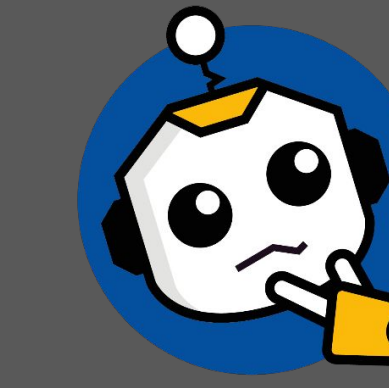
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TL;DR

Visual SLAM (VSLAM) gives globally consistent poses, key for robotic autonomy. Bundle Adjustment (BA) is typically used in VSLAM, but its cubic complexity with number of poses scales poorly to long trajectories.

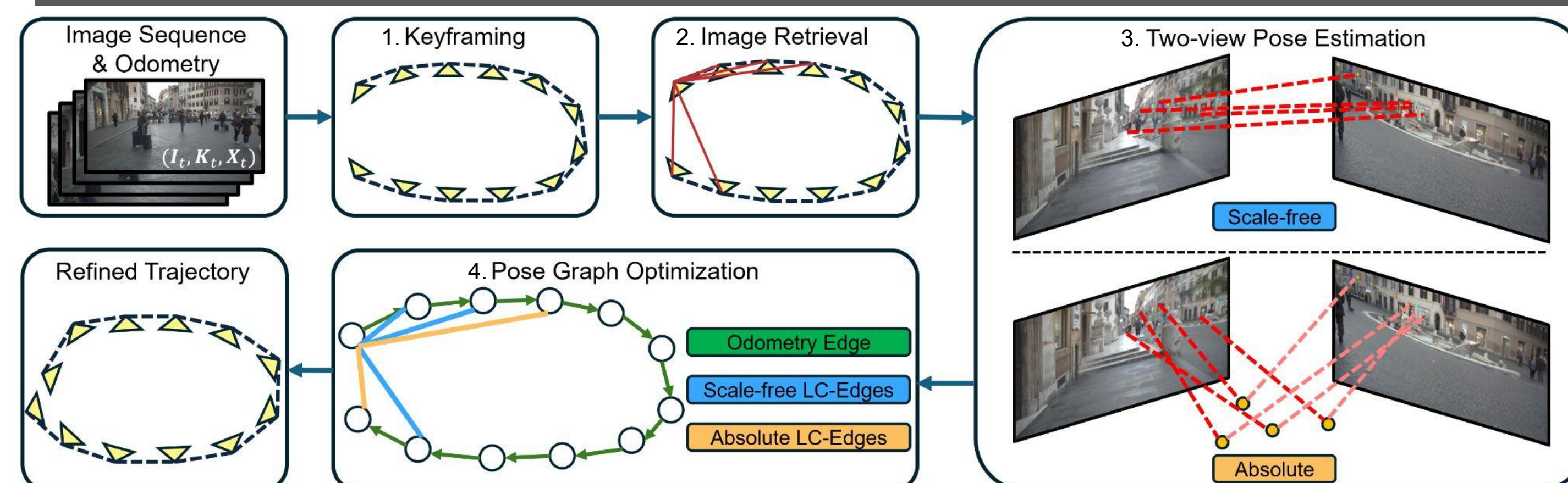
In contrast, our reconstruction-free VSLAM approach, “**2GO**”, refines trajectories **using only two-view loop closures**.

Leveraging advances in Visual Place Recognition (VPR), image matching, and Monocular Depth Estimation (MDE), 2GO:

- supports **real-time performance**,
- **scales well in map size and trajectory duration**,
- and has **competitive trajectory accuracy**.

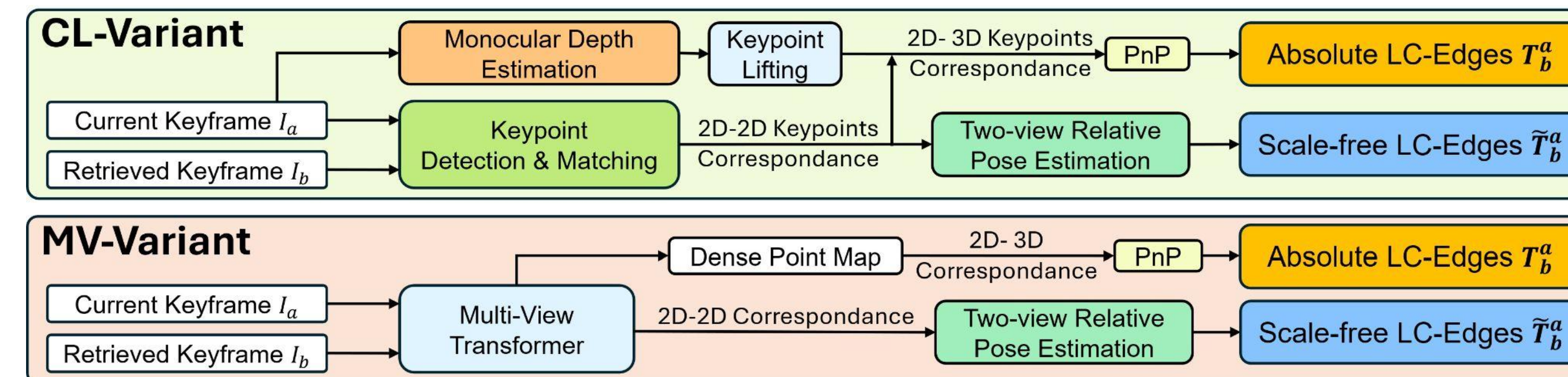
2GO effectively advances **VSLAM for long-duration deployments in large environments**.

System Overview



2GO takes a stream of **posed, calibrated** images as input.

1. **Keyframing**, based on distance from previous keyframe.
2. **Image Retrieval** identifies Loop Closure (LC) candidates via:
 - a) **Visual Similarity**: Global descriptors are computed with a VPR model. Prev keyframes with high similarity are retrieved.
 - b) **Spatial Proximity**: Prev keyframes which are spatially close are retrieved.
3. **Two-view Pose Estimation** establishes LC-Edges -- pose constraints between the current and previous keyframes. Both up-to-scale “Scale-Free” and metric-scale “Absolute” LC-Edges are computed.
4. **Pose Graph Optimisation** (PGO) robustly optimises odometry and LC-Edges to refine the trajectory.



Two-view Pose Estimation

The relative pose between two views is only known up-to-scale. Along with these constraints as **Scale-free LC-Edges**, 2GO investigates two variants, which obtain a metric-scaled pose as **Absolute LC-Edges** between the two views.

CL (Correspond-Lift) finds *correspondences* between views, then uses metric MDE to *lift* keypoints from one view into 3D.

MV (Multi-View) uses a multi-view model (e.g. *MASt3R*) to jointly predict both 2D keypoint matches and 2D-3D correspondences through a single model.

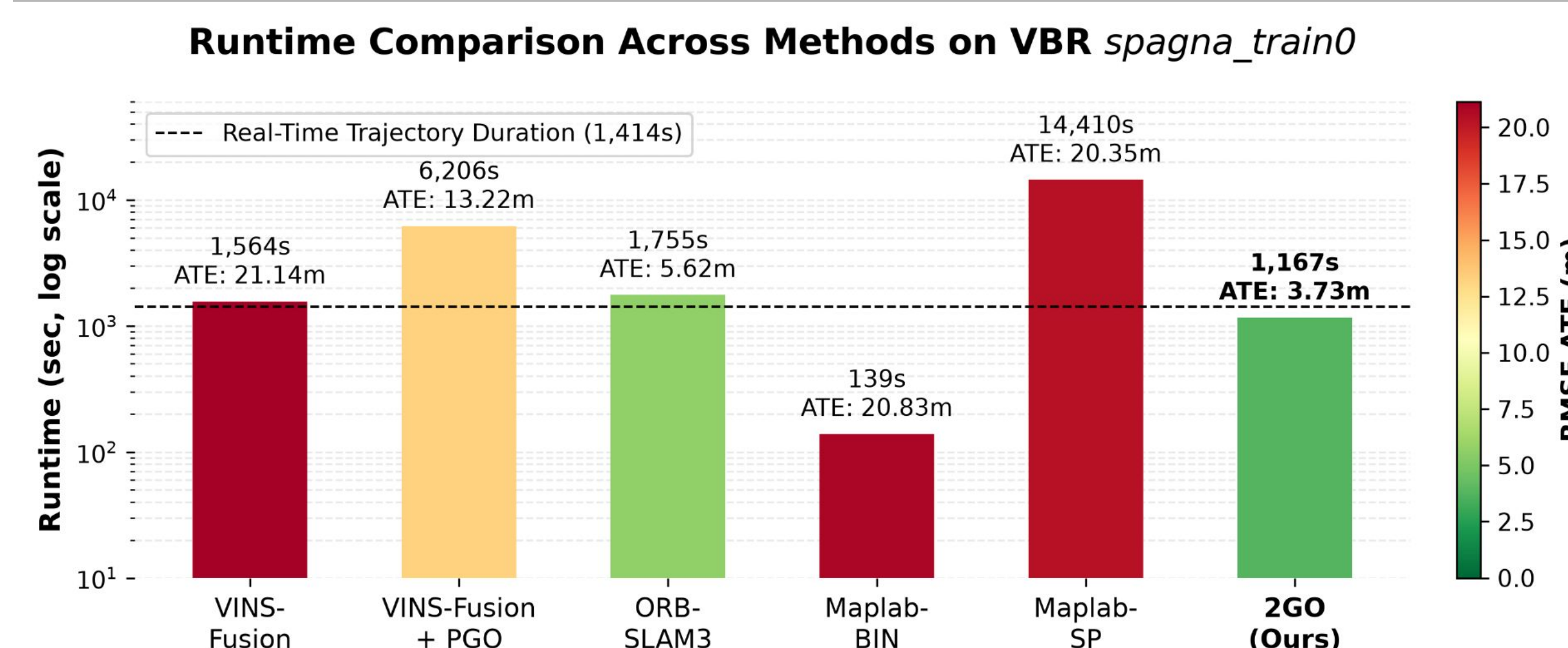
Subsequently, two-view relative pose estimation between 2D-2D correspondences yields Scale-free LC-Edges, and PnP between 2D-3D correspondences provides Absolute LC-Edges.

Experiments

2GO is evaluated on *KITTI* and *Vision Benchmark in Rome* (VBR) datasets, including trajectories in excess of 20 min.

We compare against classical VSLAM run in **stereo-only** mode, and *Maplab 2.0*, a multi-sensor mapping framework.

Scalability



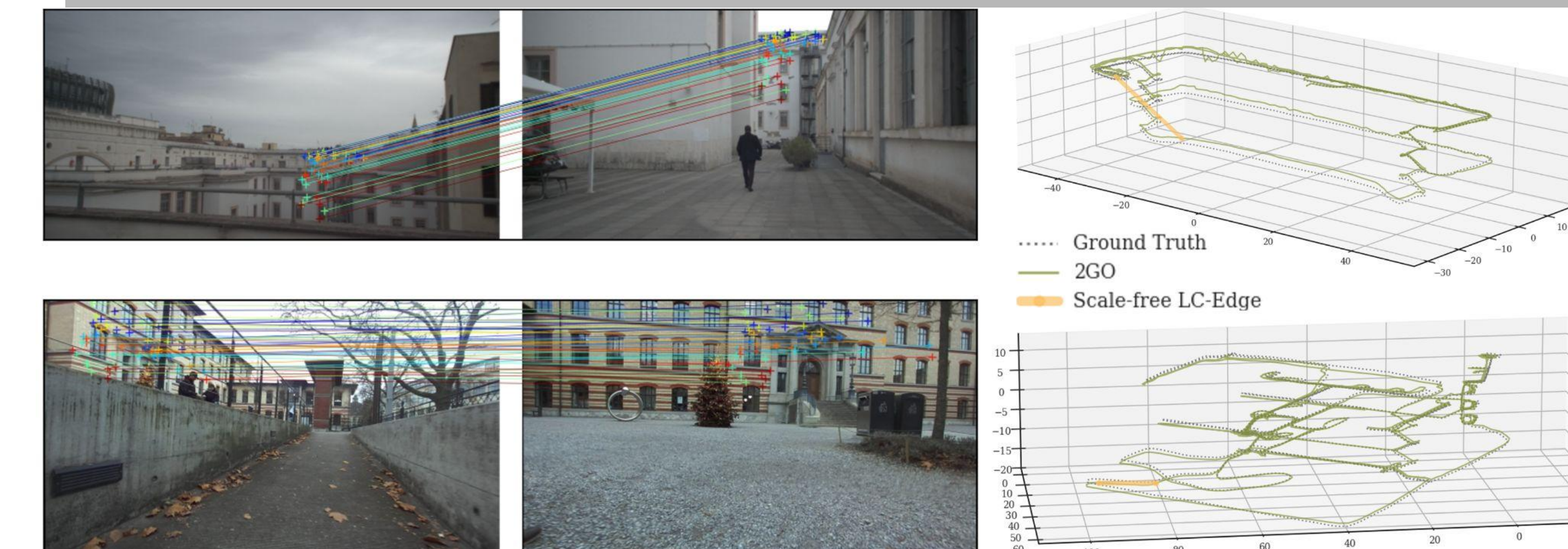
2GO is both scalable and accurate to long-duration trajectories.

Overall Accuracy

Mean RMSE ATE	VINS-Fusion	VINS-Fusion + PGO	ORB-SLAM3	Maplab-BIN	Maplab-SP	2GO (Ours)
KTITI	7.18	5.04	3.29	16.55	11.55	3.31
VBR (Driving)	68.31	49.47	25.47	66.25	63.23	14.73
VBR (Handheld)	20.32	16.86	11.61	20.17	20.07	9.90

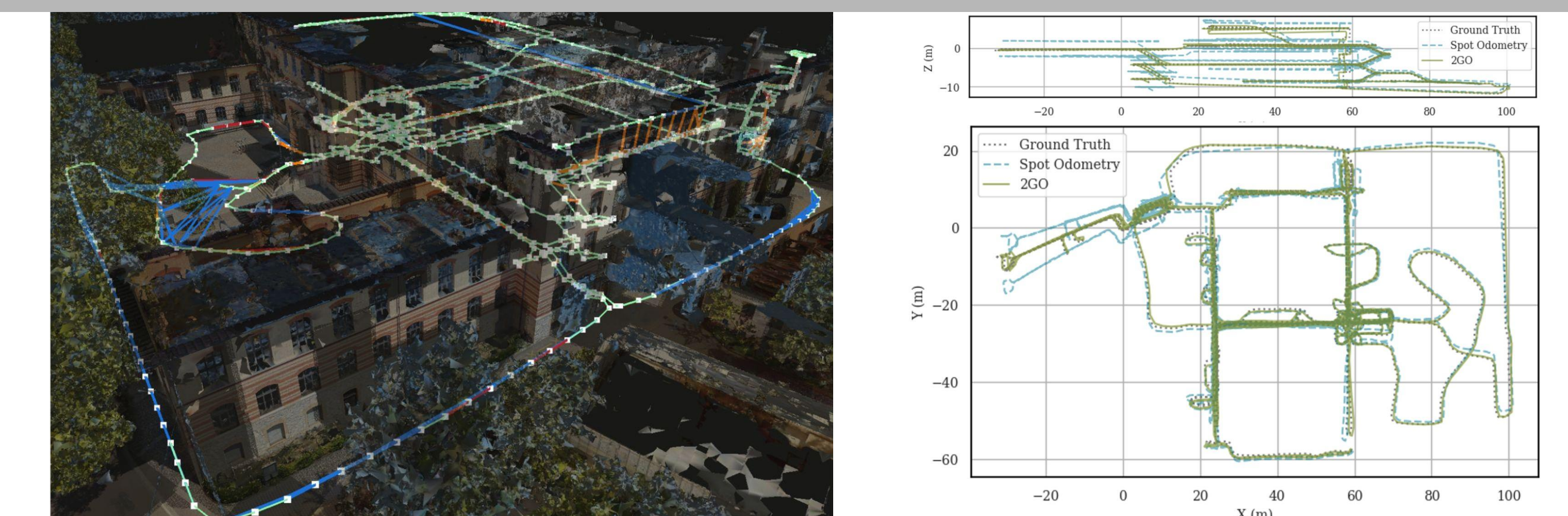
2GO performs best in mean accuracy on these datasets.

Qualitative Illustrations



Using SoTA two-view methods, 2GO incorporates loop closure information even from extreme viewpoint differences.

Real-World Evaluation



On a **1.1hr, 2.7km** recording with indoor-outdoor, multi-floor environments, 2GO requires only **15.0 min** for optimisation and reduces the input odometry's RMSE ATE from **2.44m to 0.66m**.

Conclusion and Outlook

2GO introduces a **map-free approach** to **global trajectory consistency**, using only two-view loop closures.

As two-view methods keep improving, 2GO's **modularity** allows components (e.g. VPR, MDE) to be **easily upgraded**.

2GO's two-view architecture is **deployable on resource-constrained systems**, especially those with limited GPU resources.